

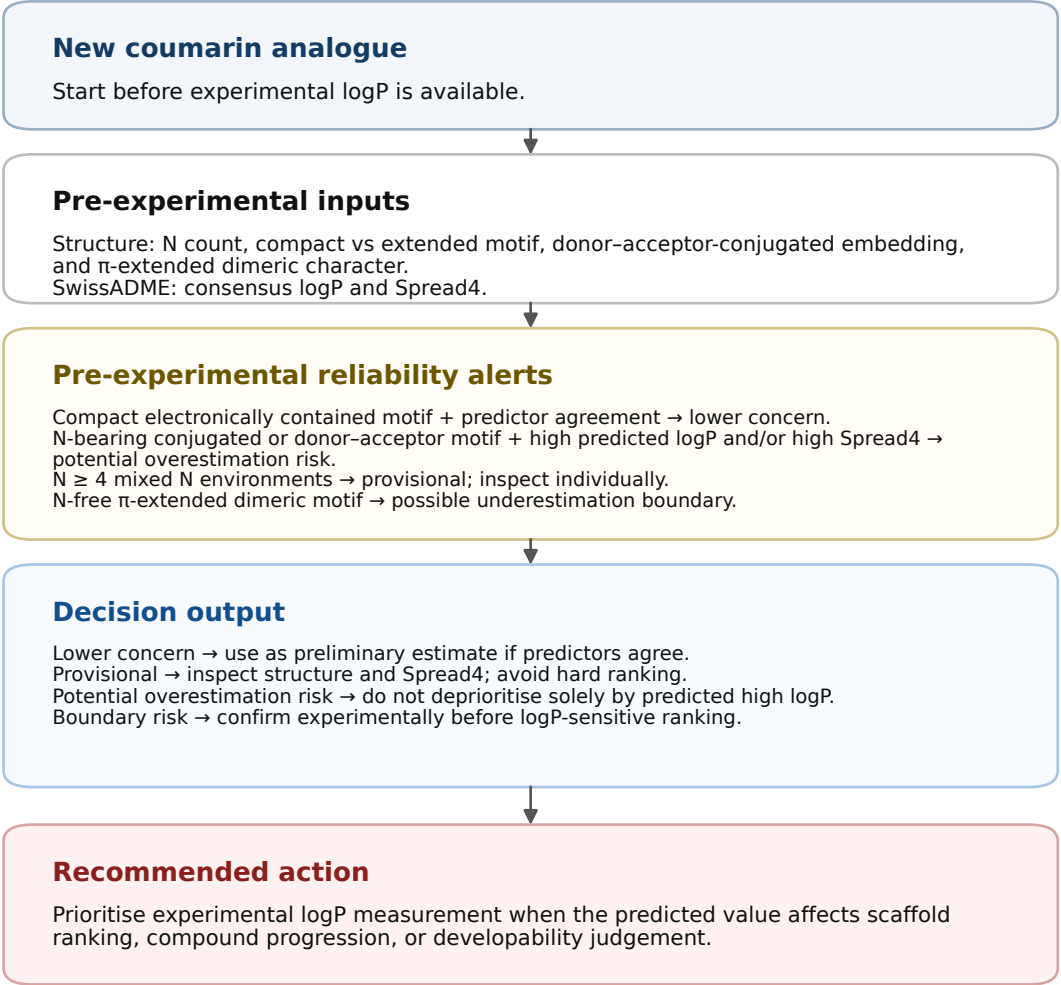
# Retrospective FM0-FM4 failure-mode taxonomy and pre-experimental reliability-triage workflow for coumarin logP prediction

Panel A is retrospective and dataset-derived; Panel B provides pre-experimental reliability alerts for new coumarin analogues.

## A Retrospective evidence base

|                                                                                                                                                                                                                                                                                                                  |                                                        |                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|------------------------------------|
| <b>Dataset-derived evidence</b><br>95-compound benchmark dataset<br>Nitrogen-dependent bias reversal<br>N-count × logP high-risk map<br>Spread4 predictor-disagreement signal<br>Documentation-tier sensitivity<br>Outlier-robustness analysis<br>External comparator audit<br>Targeted DFT/MEP diagnostic panel |                                                        |                                    |
| <b>FM0</b><br>Serviceable reference                                                                                                                                                                                                                                                                              | N = 0, exp. logP ≤ 3.0<br>n = 6; mean bias = -0.26     | Use cautiously if predictors agree |
| <b>FM1</b><br>Principal polar overestimation                                                                                                                                                                                                                                                                     | N = 1–3, exp. logP < 1.5<br>n = 13; mean bias = -2.03  | Measure experimentally             |
| <b>FM2</b><br>Conjugated-N misassignment                                                                                                                                                                                                                                                                         | N = 1–3, extended π<br>n = 44; mean bias = -0.87       | Provisional; check Spread4         |
| <b>FM3</b><br>Mixed high-N regime                                                                                                                                                                                                                                                                                | N ≥ 4, mixed environments<br>n = 27; mean bias = -0.78 | Inspect individually               |
| <b>FM4</b><br>N-free π-extended boundary                                                                                                                                                                                                                                                                         | N = 0, exp. logP > 3.0<br>n = 5; mean bias = +0.75     | Confirm when logP matters          |

## B Pre-experimental reliability-triage workflow



Pre-experimental alerts do not assign FM1 directly: FM1 requires experimental logP. This is a measurement-priority workflow, not a correction equation, replacement for experiment, or trained classifier.